Pergamon

Available online at www.sciencedirect.com

SCIENCE DIRECT®

Acta Materialia 51 (2003) 373–386

Acta MATERIALIA

www.actamat-journals.com

Study of the cold compaction of composite powders by the discrete element method

C.L. Martin *, D. Bouvard

Laboratoire GPM2, Institut National Polytechnique de Grenoble, UMR CNRS 5010, ENSPG, BP46, 38402 Saint Martin d'Hères cedex, France

Received 5 April 2002; received in revised form 5 September 2002; accepted 9 September 2002

Abstract

We present numerical simulations of the cold isostatic and close die compaction of mixtures of soft and hard powders. The simulations use the discrete element method with periodic boundary conditions on packing of 4000 spherical monosize particles. The two mechanisms that are generally put forward to explain the retarding effect of hard particles on the compaction have been analysed. First, we have studied in which condition the hard particles form a cluster that hinders the homogeneous distribution of the load. Friction between particles is shown to affect significantly the formation of such a cluster. Second, the additional deformation that soft particles must undergo in the presence of hard non-deformable particles has been evaluated. Related to this last issue, the importance of including hardening effects in the constitutive equation of the soft phase is demonstrated.

Keywords: Powder consolidation; Discrete element method; Composites

1. Introduction

Consolidation of powder composites is a typical process route to obtain metal matrix composites with high volume fractions of hard phase. In general, powder compacts are obtained via cold or hot uniaxial compaction or hydrostatic pressing in a first step. A secondary step may involve pressureless sintering or severe plastic deformation at high temperature depending on the final application. There is a large body of literature concerning the densification mechanisms of powder compacts containing hard particles during cold or hot pressing. Lange et al. [1] have studied for example a model system of hard and soft spheres with different size ratios. These authors suggest two main mechanisms to explain the cold compaction behavior of composite powders. They propose that the densification of such a system, compared to a soft powder alone, should be inhibited by the presence of an "excluded" volume around the hard powder particles. Indeed, the soft particles must undergo additional deformation in order to fill this volume partially. Another reason for the smaller densification of composites is that the load may not

* Corresponding author: Tel.: +33-4-76-82-63-37; fax: +33-4-76-82-63-82.

E-mail address: christophe.martin@gpm2.inpg.fr (C.L. Martin).

1359-6454/03/$30.00 © 2002 Acta Materialia Inc. Published by Elsevier Science Ltd. All rights reserved.
doi:10.1016/S1359-6454(02)00402-0

be entirely transmitted to the soft particles because of the formation of a continuous network of hard particles that supports a portion of the load. In order to elucidate the transmission of pressure between particles in composites, Bouvard and Lange [2] have proposed an analytical approach combined with numerical simulations of the percolation of the hard phase particles as a function of the hard to soft particle size ratio and fraction. In accordance with experimental investigations [3] it has been numerically confirmed that densification is more difficult for powders with a small hard to soft particle size ratio (see [4] for example). Also, Zavaliangos and Laptev [5] have demonstrated the importance of the relative rearrangement of particles during the compaction of composites. These authors have shown that pressure cycling can greatly enhance consolidation of composite powders. Their study confirms the importance of superimposing local deviatoric stresses to an overall isostatic stress state to help plastic deformation of soft particles and local rearrangement of hard particles.

Modeling of the compaction of powder composites has been conducted mainly by using the micromechanical approach. In that case a constitutive equation giving the contact force as a function of the local indentation between two particles is needed, together with a methodology to relate the local kinematics of the particles to the macroscopic strain field. Additionally an averaging technique is used to derive the macroscopic stress from the distribution of local contact forces. Generalizing the approach of Artz [6], Bouvard [7] proposed a model specialized to the case of hot isostatic compaction of composite powders for which rate dependence is dominant for the behavior of soft particles. Zavaliangos et al. [8] have developed a similar model but with a different force transmission assumption for the contacts between hard particles. Also, Storåkers et al. [9] have extended their work on homogeneous compacts to account for the presence of hard particles. Their approach permits the treatment of two populations of particles with different sizes and different mechanical properties (with the possibility of strain and strain-rate dependence). These three approaches [7–9] bear some important common features. In particular, they have in common that they depart from the approach generally adopted for a homogeneous compact [10] through the kinematics adopted for the local indentation between particles. Since hard and soft particles indent each other differently, the local indentation between particles cannot be directly calculated from the macroscopic strain field. In absence of this information, the force distribution in the compact cannot be derived from the macroscopic strain field. Thus an additional assumption on the force distribution must be used. A natural assumption is that all contact forces, irrespective of their nature (hard or soft) are equal for a given orientation [7,9]. This assumption seems valid if hard phase particles are not numerous enough in the composite to shield soft particles or if they are able to rearrange sufficiently. However if clusters of hard particles form and carry a substantial part of the load, the contact forces may vary largely between soft and hard particles. The limit between these two situations depends on the type of arrangement between particles and on the nature of the contacts between particles. Using a truss model on a ordered crystalline compact with no sliding possibility between the particles, Zavaliangos et al. [8] have observed that the load may vary largely from hard to soft particles and used their numerical results in their model. Kim et al. [11] have studied the effect on the macroscopic stress response of using an equal contact force between particles as in [9] and different contact forces as proposed in [8]. They have demonstrated that the force distribution determined by Zavaliangos et al. [8] leads to much larger stress than in the case of equal forces between all particles.

Jagota and Scherer [12,13] have shown that the discrete element method (DEM) for which each particle of an assembly is modelled separately should be able to provide some more insight into the densification of composites. The method was used on a compact of particles submitted to densification with a linear viscous law at the contact. They have shown in particular that for a 3D randomly packed structure the limit fraction of hard particles above which the apparent viscosity of the packing increases drastically depends predominantly on the nature of the contact between the hard particles [13]. If the hard particles are not allowed to slide with each other or to rotate, this threshold coincides with the minimal volumetric fraction leading to percolation for a randomly packed structure (0.32). However, if less kinematic constraints are imposed on the contacts between hard particles, then the threshold may be much larger and may even approach unity when sliding and rotating of hard particles are allowed. These numerical observations are in qualitative accordance with experimental results on the cold compaction of mixtures of soft and hard particles that are sliding at ambient temperature [1,3,11,14,15]. The compaction curves collected by these authors do not show any sign of mechanical percolation even for large fractions of hard particles (up to 60%). This set of results shows that it is important to take full account of the rearrangement occurring between particles during the compaction of composites and to account for possible sliding between hard particles. This cannot be properly done with analytical models [7–9,11] but can be achieved with the DEM. Using this method on homogeneous packing, we have already shown that the introduction of rearrangement affects the average number of contacts between particles, their contact areas and the distribution of these quantities in the packing [16]. In any case, rearrangement has a softening effect on the stress response of the compact. Concerning mixtures of soft and hard particles, Skrinjar and Larsson [17] have recently carried out simulations using the DEM. These authors limited their analysis to an ideally plastic soft phase in isostatic conditions and with frictionless contacts. Also, the low initial relative density attained in their simulations does allow a direct comparison with analytical and experimental results.

In that context, the present study uses the methodology reported in [16] to analyse in a detailed manner the compaction of 3D packings for mixtures of soft and hard particles. The general aim is to study the effect of rearrangement on the compaction behaviour of such composites. We briefly describe the numerical model in Section 2. Section 3 presents some results on the evolution of coordination number in these composites as compaction proceeds and compares the results with existing analytical expressions. Section 4 discusses how the contact tangential law affects the distribution of contact forces in the compact, in particular whether all contact forces between soft and hard particles can be assumed equal or not as discussed above. The macroscopic stress response is analysed in Section 5 both in isotropic and in close die conditions.

2. Model

2.1. Contact behaviour

The two populations of particles ($p = 1,2$) are modelled as spheres made of an elasto-plastic material with elastic constants E_p and ν_p and the following strain hardening plastic law:

$$\sigma_e = \Sigma_p \varepsilon_e^{\frac{1}{m}}, \qquad p = 1,2 \tag{1}$$

where ν_p is a material constant for particle of type p, m is the hardening coefficient and σ_e and ϵ_e are the stress and strain in the uniaxial case. Note that m is the same for both materials. For a contact between contacting particles of type p and q with radius r_p and r_q (Fig. 1), we note:

$$r_{pq} = \frac{r_p r_q}{r_p + r_q}, \qquad p,q = 1,2 \tag{2}$$

$$E_{pq} = \frac{4}{3}\left(\frac{1-\nu_p^2}{E_p} + \frac{1-\nu_q^2}{E_q}\right)^{-1}, \qquad p,q = 1,2 \tag{3}$$

$$\Sigma_{pq} = (\Sigma_p^{-m} + \Sigma_q^{-m})^{-\frac{1}{m}}, \qquad p,q = 1,2 \tag{4}$$

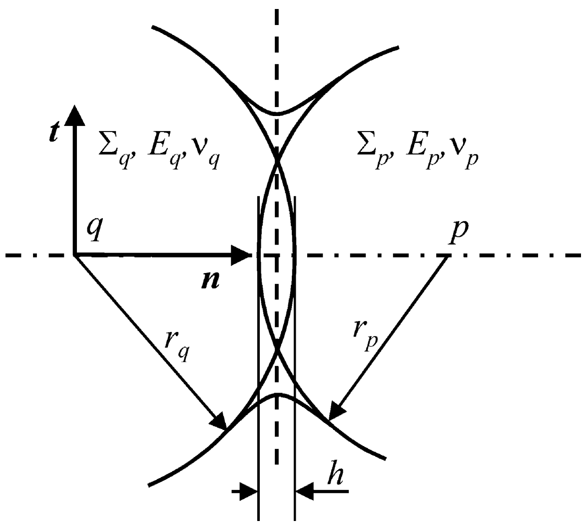

Fig. 1. Two-dimensional sketch of the contact geometry between two spherical particles p and q.

Particles overlap with each other during compaction, with h defining the overlap (Fig. 1). The normal force at the contact is:

$$\boldsymbol{N} = -\min(N_E,N_P)\boldsymbol{n} \tag{5}$$

where $\boldsymbol{n}$ denotes the unit normal vector to the contact plane and N_E and N_P are the elastic and plastic normal contact forces. For a Hertzian contact the elastic force N_E is given by:

$$N_E = E_{pq}\sqrt{r_{pq}}\,h^{\frac{3}{2}}, \qquad p,q = 1,2 \tag{6}$$

We use the expression developed by Storåkers et al. [18] from a similarity solution for the frictionless normal plastic indentation of two dissimilar spherical particles undergoing viscoplastic deformation. In the case of rate-independent plasticity, for two spheres with a constitutive equation given by Eq. (1), the normal plastic contact force N_P is written as $(p,q = 1,2)$

$$N_P = \pi\Sigma_{pq}2^{1-\frac{1}{m}}3^{1-\frac{1}{m}}c(m)^{2+\frac{1}{m}}r_{pq}^{1-\frac{1}{2m}}h^{1+\frac{1}{2m}}, \tag{7}$$

where $c(m)$ relates the contact area A to the indentation h:

$$A = 2\pi\, c(m)^2 r_{pq}h, \qquad p,q = 1,2 \tag{8}$$

The function $c(m)^2$ has been tabulated by Storåkers et al. [18], it varies from 0.5 for linear hardening ($m = 1$) to 1.45 for ideal plasticity ($m = \infty$). The expression of N_P is only valid for frictionless contact and normal indentation but should remain approximately correct for contact with friction [19] and obliquity [20]. Also, we consider that the material transported away by plastic flow from the contact does not interact with neighbouring particles. Hence particles are considered as spheres that are simply truncated at the contact as compaction proceeds. This geometrical assumption limits the domain of validity of the simulations to the domain of relative density for which contact impingement is negligible. Also, the possibility that two neighboring contacts cannot be considered as mechanically independent limits the validity of Eqs. (7) and (8). The calculations that will be presented in the following have been conducted for relative density of up to 0.95 for sake of completeness. However, it should be clear that the geometrical restrictions and the mechanical limitations listed above may apply as early as 0.85–0.90. Hence, the underlying assumptions used in the present simulations should be considered strictly valid only under this limit.

Concerning the tangential force at the contact, we simply write that the contact is either in a sticking state, with negligible relative tangential displacement, or in a state of gross sliding. The sticking case is ensured by giving a sufficiently large tangential stiffness to the contact. When the contact is sliding, a simple Coulomb law of friction is used:

$$\boldsymbol{T} = -\mu_{pq}|\boldsymbol{N}|\boldsymbol{t}, \qquad p,q = 1,2 \tag{9}$$

where $\boldsymbol{t}$ is the unit vector parallel to the contact plane in the direction of the relative velocity. μ_{pq} is a constant effective friction coefficient for the p–q contact. A more elaborated tangential contact law that includes the effect of the obliquity angle may be introduced when the contact involves plasticity. However, we have shown that a simple constant friction law at the contact should capture correctly the relative tangential shift between two contacting particles [16].

In the following we limit the study to a mixture of a population of soft particles (population 1) with hard particles (population 2) with material constants chosen so that a contact between two hard particles stays elastic all along the compaction. The two populations are chosen with the same size, hence no size effect is introduced in the simulations. The resulting stresses from the simulations are normalised by the plastic stress parameter Σ_1 of the soft phase, which has been introduced in Eq. (1). The ratio of material properties that have been chosen for the two phases are : $E_2/E_1 = 10$, $\nu_2/\nu_1 = 1$, and $\Sigma_2/\Sigma_1 = 10^4$. The elastic properties of the soft phase are: $E_1 = 120$ GPa and $\nu_1 = 0.34$. We have verified that the modification of the elastic and plastic properties of the hard particles does not affect significantly the results of the simulations providing the ratios E_2/E_1 and Σ_2/Σ_1 are sufficiently large. For instance we have used a ratio $E_2/E_1 = 100$ for a series of simulations. These calculations are much more demanding in terms of CPU time since the time step is roughly divided by ten (as detailed below). In that case we observe that the macroscopic stress is increased by no more than 3% compared to the standard ratio $E_2/E_1 = 10$ used in all other simulations. This means that the elastic strain of the hard phase particles is negligible and does not contribute to the overall densification of the composite. Note that the large Σ_2 value has no other practical meaning than to ensure that the hard-hard contacts stay elastic and that the stiffness of soft-hard contacts is $2^{1/m}$ times the stiffness of soft–soft contacts as seen by evaluation of Eqs. (4) and (7).

The rotation of particles is not included in the model. Hence the equilibrium of moment is not ensured once friction is included. The rationale for not including rotations is that the size of contacts involving plastic deformation should hinder rotations of both soft and hard particles. Also, at this stage of compaction there is a rather high coordination number (between 6 and 10 as densification proceeds) that should provide an additional barrier to rotation. Furthermore two studies using DEM on 2D packings support our view that rotations can be omitted in our problem. For contacts involving plasticity, Redanz and Fleck [21] using DEM on 2D packings have observed that the extent of rearrangement is fairly insensitive to the degree of particle rolling. They have compared their numerical results obtained with and without rotations concerning stress, coordination number and tangential rearrangement and have concluded that the rearrangement due to rotations is negligible once plasticity is involved. Since no rotational stiffness is used in their simulations, the contribution to rearrangement due to particle rotations should give an upper bound compared to the real situation for which plastic contacts of finite size hinder rotation. For linearly viscous contacts Jagota and Scherer [12] have studied the difference between a truss model (reminiscent of the present study for which only force equilibrium is enforced) and a beam model (for which both force and moment equilibrium are enforced). Their results show that the truss and beam models are equivalent concerning the behaviour of the mixture. Note that rotations would have to be included if the initial relative density were much smaller than the random close packed relative density (0.637) attained here. In that case, particles involving a small number of contacts (less than 3 or 4) would certainly be able to rearrange by rotations. Similarly, for packings with a very large volume fraction of hard particles involving elastic contacts, rearrangement would operate both by relative sliding and rotation.

2.2. Discrete element method

We briefly summarise the methodology adopted for the DEM. More details can be found in [16]. The interactions of the spherical particles are accounted for by modelling the evolution of the packing as a dynamic process. At each incremental time step, the contacts between particles are determined and the interparticle forces at the contact are calculated. The total force and acceleration acting on a particle are then determined and an explicit scheme enables the computation of the new position of each particle using a small time step Δt. The evaluation of the correct time step for the particles assembly is done by calculating the time step for each type of possible contact as proposed by Cundall and Strack [22]

$$\Delta \mathrm{t} = 2f_t\sqrt{\frac{M}{k_{pq}}} \tag{10}$$

where f_t is less than unity in any case to ensure calculus stability, M is the smallest particle mass of the pair of particles and k_{pq} is the normal contact stiffness (defined as $\mathrm{d}|\boldsymbol{N}|/\mathrm{d}h$). For a contact between two soft particles or a soft and a hard particle, the contact stiffness is given by the approximate expression (for sufficiently large values of m):

$$k_{1q} \approx \pi\Sigma_{1q}2^{1-\frac{1}{m}}3^{1-\frac{1}{m}}c(m)^{2+\frac{1}{m}}r_{1q}\ , \ q = 1,2 \tag{11}$$

For a contact between two elastic hard particles the contact stiffness is:

$$k_{22} = E_{22}\sqrt{r_{22}h_{22}} \tag{12}$$

where h_{22} is the indentation between two hard particles in the packing. Depending on the material constants and the state of compaction, the time step may be imposed either by plastic contacts involving a soft particle (Eq. (11)) or by elastic contacts between hard particles (Eq. (12)). We upscale the nominal density of the particles in order to enlarge the time step to reasonable values of the order of

10^{-3} s (Eq. (10)) [23]. Upscaling the nominal density of the particles causes the accelerations and velocities to be reduced by orders of magnitude while forces and strains remain unchanged. The equilibrium positions of the particles are not affected by the value of the upscaling factor if an adequately small strain rate is imposed (10^{-4} s^{-1}), and checks are included to ensure that each particle is sufficiently close to equilibrium.

The boundary conditions on the packing are given by periodic conditions in a representative cubic cell. The macroscopic rate of deformation of the cell $\dot{\varepsilon}_{ij}$ is specified by imposing at the beginning of the time step a translation

$$\Delta x_i = \dot{\varepsilon}_{ij} x_j \Delta t \tag{13}$$

to each particle centre. Similarly the faces of the cell are translated accordingly. The displacements calculated by taking into account the interaction between particles are summed with these imposed translations to obtain the final position of the particles at the end of the time step. The macroscopic stress tensor is calculated by using the following expression [23,24]

$$\sigma_{ij} = \frac{1}{V}\Big(\sum_c (r_p + r_q - h)\, N n_i . n_j + \sum_c (r_p + r_q - h)\, T n_i . t_j\Big) \tag{14}$$

where V is the volume of the sample, and the summation is carried out on all contacts.

We note f_p the fraction of particles of type p relative to the total volume of solids and we define the volume fraction of particles of type p relative to the total volume of the compact:

$$\phi_p = f_p D\,, \qquad p = 1{,}2 \tag{15}$$

where D is the relative density. The total number of particles is 4000. Initial randomly packed samples have been prepared with four different amounts of hard particles: $f_2 = 10$, 20, 30 and 40%. The preparation method consists first in randomly replacing soft particles by hard particles in previously homogeneous samples as prepared in [16]. The particles are then left to rearrange according to their new material properties in order to attain local equilibrium with the condition that only one contact in the compact is plastic. The relative density of the initial composite packings obtained with this condition is not strongly affected by the presence of hard particles and is 0.637 as for the homogeneous packing [16]. Similarly the initial average coordination number is not modified and is approximately 6.2 for the composites as for the homogeneous packing. The initial configuration of particles is shown in Fig. 2 for 30% of hard particles together with the packing after isostatic compaction at 0.90 relative density.

3. Coordination number

We denote Z as the average coordination number of the packing, Z_{11} as the average coordination of soft particles with each other, Z_{12} as the average coordination of soft particles with hard particles, and so on. For monosize isotropic mixtures of homogeneous particles, a common hypothesis is that the average number of contacts of type p particles with type q particles is proportional to the average coordination number Z and to the fraction of type q particles [2,25] with:

$$Z_{pq} = f_q Z\,, \qquad p,q = 1{,}2 \tag{16}$$

The increase of the average coordination number with relative density has been calculated by analytical approaches that consist of considering the concentric growth of a particle in an average Voronoi cell [6] or equivalently by reducing the centre to centre spacing between two representative particles [10]:

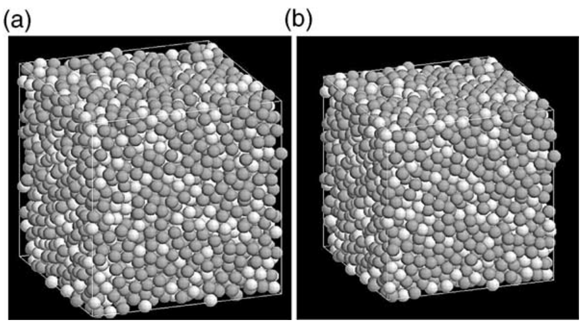

Fig. 2. (a) Initial configuration of the packing containing 30% of hard particles ($D = 0.64$). (b) The same packing at 0.90 relative density. Hard particles are white and soft particles are grey.

$$Z = Z_0 + C\left(\left(\frac{D}{D_0}\right)^{1/3} - 1\right) \quad (17)$$

where D_0 is 0.64, $Z_0 = 7.3$ and $C = 15.5$ is taken from the experimental data of Mason [27]. It should be pointed out that Eq. (17) is valid only when the material at the contact does not interact with neighbouring particles (short-range redistribution). The same assumption is adopted for the DEM simulations presented here. This assumption is valid for cold compaction whereas at higher temperature, long-range redistribution seems more appropriate. In the case of long-range redistribution assumption, Artz [6] has proposed another formulation that has been fitted by Helle et al. [26] with the following empirical equation:

$$Z = 12D \quad (18)$$

Eqs. (17) and (18) have been used in analytical models [7–9,11] for modelling the evolution of contact number in composite powders. However, Eqs. (16)–(18) are not specifically designed for mixtures of soft and hard particles, thus DEM simulations provide a way to verify that they hold for the cold compaction of such composites. We carry out this part of the study by simply assuming an ideally plastic material for the soft particles.

First we present the evolution of Z for a homogeneous packing and for composites, for which all contacts are set with a 0.1 friction coefficient. Fig. 3 shows that the average coordination number in the composite does not depend on the fraction of hard particles and follows the same curve as for the homogeneous packing. The same trend is observed for the case where only soft-soft contacts are set with some friction and hard-hard contacts are frictionless. The only difference is that the average coordination number is slightly higher owing to the fact that more local rearrangement is taking place compared to the case where all contacts see friction. Apart from the region of very low relative density, our simulations yield coordination values that are consistent with Eqs. (17) and (18). Recall that Eq. (17) has been derived by assuming that no rearrangement is taking place [6,10] whereas local rearrangement which is taken into account in our simulations tend to increase the coordination number [16]. The initial coordination number $Z_0 = 7.3$ used in Eq. (17) is taken from a linear fit to the cumulative radial distribution function determined by Mason [27] and is quite large compared to the generally accepted value of 6.2 for randomly close packed assemblies of monosize spherical particles [28,29]. However it may prove a valid choice for practical conditions for which the powder particles are not strictly monosize nor strictly spherical.

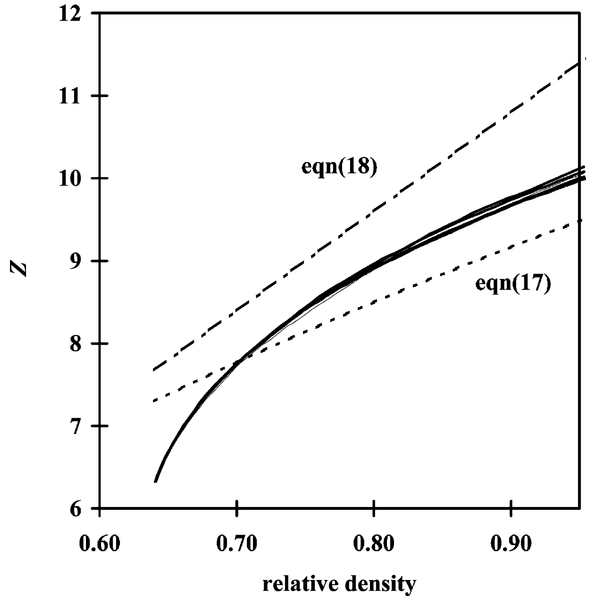

Fig. 3. Evolution of the average coordination number for $f_2 = 0$, 10, 20, 30 and 40% fraction of hard particles in the case where all contacts are set with a 0.1 friction coefficient. The presence of hard particles has no significant effect on the average coordination number. Comparison with Eq. (17) derived by Arzt [6] and Fleck [10] and with Eq. (18) proposed by Helle et al. [26].

In Fig. 4, we explore the validity of Eq. (16) that states that all the curves Z_{pq}/f_q should fall on the same master curve given by Z. Fig. 4 shows that the relation is followed quite well except for the number of hard–hard contacts for $f_2 = 30$ and 40%. The explanation for the slightly smaller value of Z_{22}/f_2 is that hard particles do not indent each other as much as soft particles since there exists an excluded volume around hard particles. Hence contacts due to the centre to centre approach of hard particles are less frequent than for soft particles. In any case the deviation from the master

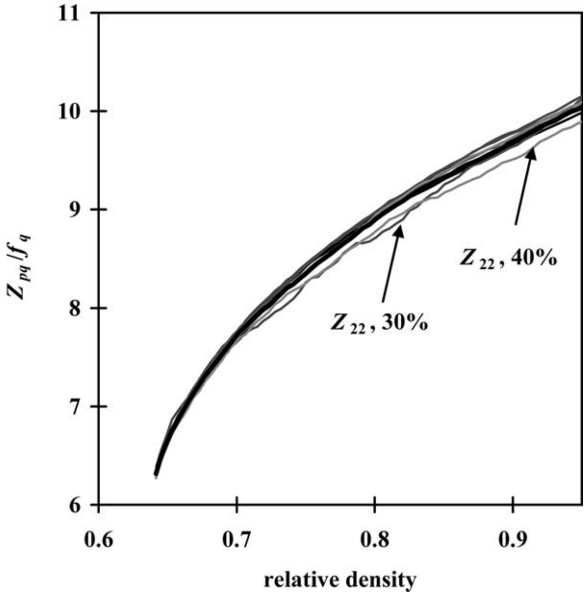

Fig. 4. Evolution of the average coordination number Z_{pq} normalised by f_q for $f_2 = 0$, 10, 20, 30 and 40% fraction of hard particles in the case where all contacts are set with a 0.1 friction coefficient. Only the Z_{22}/f_2 curves for 20 and 40% fraction of hard particles depart from the master curve.

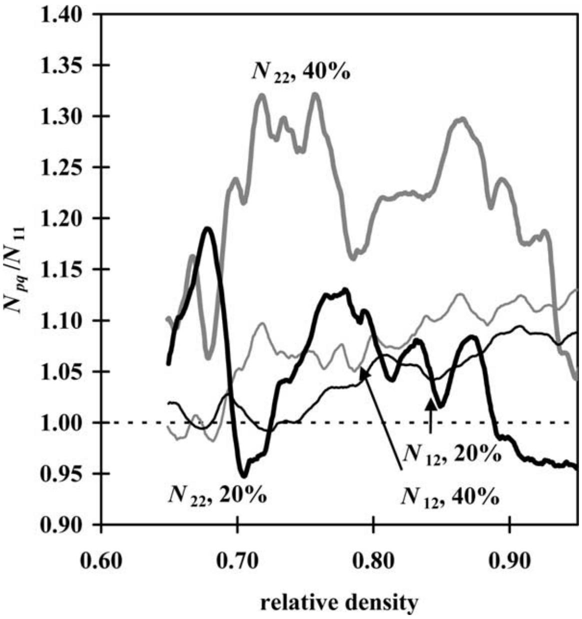

Fig. 5. Evolution of the average contact forces with relative density for frictionless contacts. The average contact forces N_{12} (soft-hard contacts) and N_{22} (hard-hard contacts) are normalised by N_{11} (average force for soft-soft contacts).

curve is rather small and Eq. (16) can be considered valid for monosize composite systems with different mechanical properties.

4. Distribution of contact forces

In this section we consider soft particles as ideally plastic ($1/m = 0$). We denote N_{11} as the average of normal contact forces for soft–soft contacts, N_{12} as the average of normal contact forces for soft–hard contacts and N_{22} as the average of normal contact forces for hard–hard contacts. First we investigate the distribution of contact forces in composites containing 20 and 40% volume of hard particles with frictionless contacts. In that case, rearrangement between particles is maximum since no kinematics constraint exist except for the normal indentation on each contact. Fig. 5 shows the evolution of the ratios N_{12}/N_{11} and N_{22}/N_{11} with relative density. We observe that no clear trend is discernible regarding the evolution of the ratios with relative density for the 20% composite while the 40% composite indicates a slight increase of the ratio N_{22}/N_{11}. The ratios are increasing with the fraction of hard particles, but in any case the values of the ratios are in the order of magnitude of unity.

When rearrangement capability of the particles is hindered by the introduction of some realistic friction coefficient between particles, the distribution of forces is modified. We have studied two cases: one where only the friction coefficient of soft–soft contacts is set to 0.1, the other where all contacts, whatever their nature, are characterized by a 0.1 friction coefficient. Fig. 6a and b assemble the results of the different simulations for the two cases studied in terms of the friction coefficient. Contrasting with the frictionless case, the ratios N_{pq}/N_{11} are increasing with Φ_2. The figures show that the value of the normal force at soft–hard contacts is not greatly affected by the friction coefficient of hard–hard contacts and that in any case the normal force at soft–hard contacts is no greater than 1.3 times the normal force at soft–soft contacts (Fig. 6a). This is due to the fact that the stiffness of a soft–hard contact is the same to that of

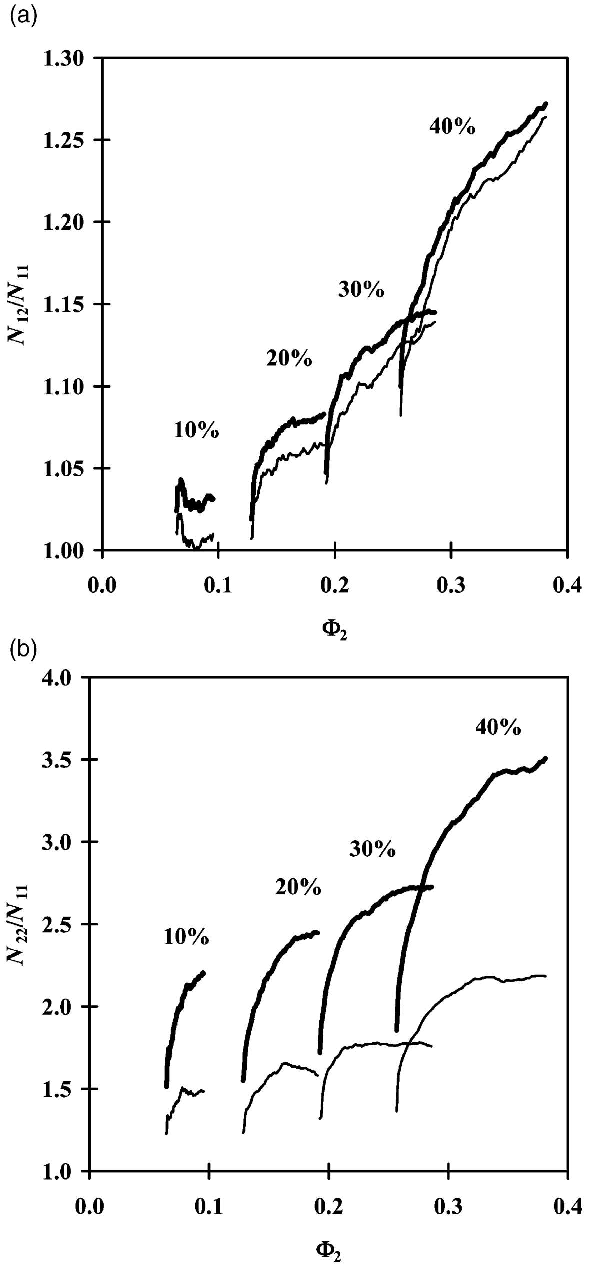

Fig. 6. Effect of the volume fraction of hard phase particles on the (a) soft–hard average force N_{12} (normalised by N_{11}) and (b) hard–hard average force N_{22} (normalised by N_{11}) for two different friction cases: only soft–soft contacts are set with a 0.1 friction coefficient (thin curves); all contacts are set with a 0.1 friction coefficient (thick curves). Φ_2 is the volume fraction of hard particles relative to the total volume of the compact (Eq. (15)).

a soft–soft contact (see Eqs. (4) and (7)) for an ideally plastic soft material. By contrast, the behavior of hard–hard contacts is affected by the introduction of friction and in the case where friction is introduced for hard–hard contacts, the ratio N_{22}/N_{11} takes value of the order of three for the highest fraction of hard particles (Fig. 6b). This is in accord with the general view that as the amount of hard particles increases to values larger than 20%, a network of hard particles forms that supports a proportionally larger share of the load compared to the soft particles. As demonstrated by our simulations with different friction coefficients at contacts, the proportion of the load that is supported by this network depends greatly on the friction coefficient or in other terms depends on the local rearrangement capability of the network of hard particles. Our results that include realistic sliding at the contact show that contact forces cannot be considered equal on average as assumed in [7,9]. However the ratio N_{22}/N_{11} given by our simulations, is much less than the value calculated by Zavaliangos et al. [8] who obtained $N_{22}/N_{11}\approx 4.4$ for 20% of hard particles and $N_{22}/N_{11}\approx 9.6$ for 40% of hard particles. The main reason for the discrepancy between our results and those of Zavaliangos et al. [8] is that the latter were conducted on a ordered crystalline compact with no sliding possibility for the particles. Hence the results presented in Fig. 6b should provide a realistic evaluation of the ratio N_{22}/N_{11} in between two extreme cases: frictionless contacts and fully adhesive contacts.

5. Macroscopic stress response

5.1. Isostatic compaction

We consider first the isostatic compaction response of a ideally plastic soft powder mixed with hard elastic particles. Fig. 7 shows the evolution of pressure as compaction proceeds in the case where all contacts are set with a 0.1 friction coefficient. The effect of hard particles is, as observed commonly, to increase the pressure necessary to obtain a given relative density. As mentioned above, the same simulations were per-

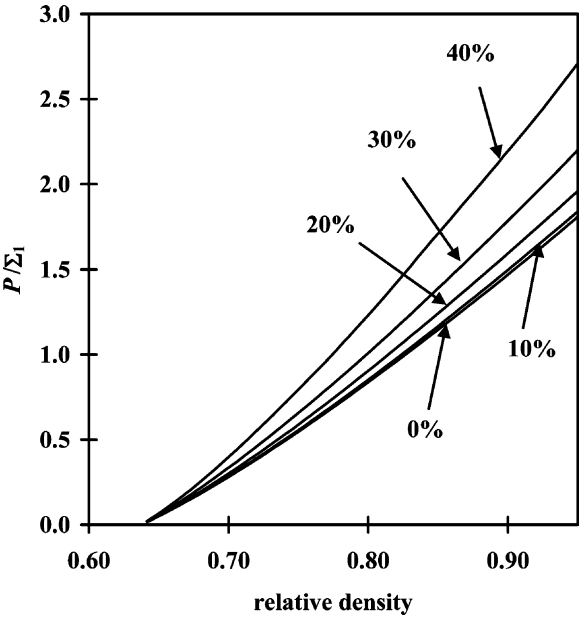

Fig. 7. Evolution of pressure for soft particles considered as ideally plastic ($1/m = 0$) in the case where all contacts are set with a 0.1 friction coefficient. The pressure is normalised by Σ_1 (Eq. (1))

formed with an elastic constant for the hard phase set to $E_2 = 100E_1$ instead of $E_2 = 10E_1$. The resulting pressure is only increased by less than 3% for the highest fraction of hard particles compared to the case $E_2 = 10E_1$. This shows that the elastic behaviour of the hard phase has only a minor effect on the compaction behaviour of the composite even for high amounts of hard phase.

The effect of the presence of hard particles on the compaction process can be captured more easily through the ratio of the pressure required to achieve a given relative density in the composite to the pressure required to achieve the same relative density for the homogeneous compact. We denote K_h as the ratio for the hydrostatic stress state. Note that since we are dealing with monosize mixtures of hard and soft particles, all the composites prepared here have approximately the same initial relative density and coordination number compared to the homogeneous compact. Hence the constraint factor K_h accounts solely for the difference in mechanical properties between the two powders.

The results on the distribution of contact forces (Section 3) have shown that the friction coefficient between particles has minor effect on the forces at soft–hard contacts and significant effect on the forces at hard–hard contacts. Predictably, Fig. 8 shows that the constraint factor K_h is itself dependent on the friction law between particles. Note that K_h is increasing as compaction proceeds and that the evolution of K_h replicates the trend of the ratio N_{22}/N_{11} shown in Fig. 6b. Hence it can be deduced that the increase of K_h during the compaction is at least partially related to the increase of the ratio N_{22}/N_{11}. The results of the DEM simulation concerning the constraint factor are in reasonable accordance with the model of Storåkers et al. [9] only for small amounts of hard phase. This is because this model assumes that $N_{22} = N_{12} = N_{11}$. This assumption is approximately verified only for small amounts of hard phase and frictionless contacts between hard particles but not for large amounts of hard phase or frictional contacts

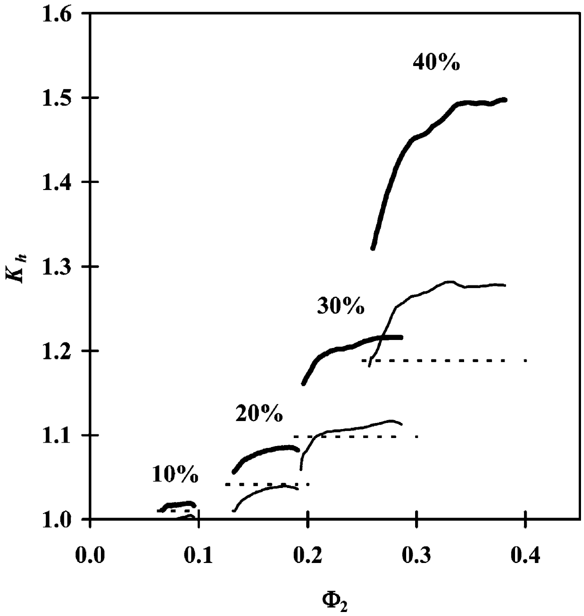

Fig. 8. Evolution of the constraint factor K_h in hydrostatic condition for soft particles considered as ideally plastic ($1/m = 0$) for two different friction cases: only soft–soft contacts are set with a 0.1 friction coefficient (thin curves); all contacts are set with a 0.1 friction coefficient (thick curves). The dotted curve represents the model of Storåkers et al. [9]. Φ_2 is the volume fraction of hard particles relative to the total volume of the compact (Eq. (15)).

between hard particles (Fig. 6). Note also that a direct comparison between the DEM simulations presented here and the model of Storåkers et al. [9] is somewhat difficult since particle rearrangement, which softens the stress response of the compact, is not taken into account in their model.

In order to investigate the role of soft particle deformation on the overall behaviour, we have introduced the possibility of strain hardening in the constitutive equations (Eqs. (1) and (7)). The addition of strain hardening permits a more realistic simulation of the soft phase contribution. Also it allows putting into relief the role of the additional deformation of the soft particles required by the presence of non-deformable particles. Indeed, Fig. 9 shows that increasing the hardening exponent $1/m$ has a significant effect on the value of K_h. We have verified that for a given relative density, soft particles in the composite deform more than in the homogeneous compact. However, soft–soft contacts and soft–hard contacts are not equally affected by hard particles: soft–soft contacts do not deform significantly more in the presence of hard particles, whereas soft–hard contacts do. For example, for f_2 = 40% the average indentation h at soft–hard contacts is 20–40% larger than the average indentation in the homogeneous material. The effect of this additional deformation on the macroscopic pressure increases with the hardening exponent, hence explaining the effect of the hardening exponent on the constraint factor. In parallel, the effect of the hardening parameter can be explained by noting that the stiffness of soft–hard contacts is larger than the stiffness of soft–soft contacts by a factor $2^{1/m}$ (Eqs. (4) and (7)). Hence, contrary to the ideally plastic case, soft–hard contacts play a significant role in the increase of the macroscopic stress for mixtures of hard and soft particles.

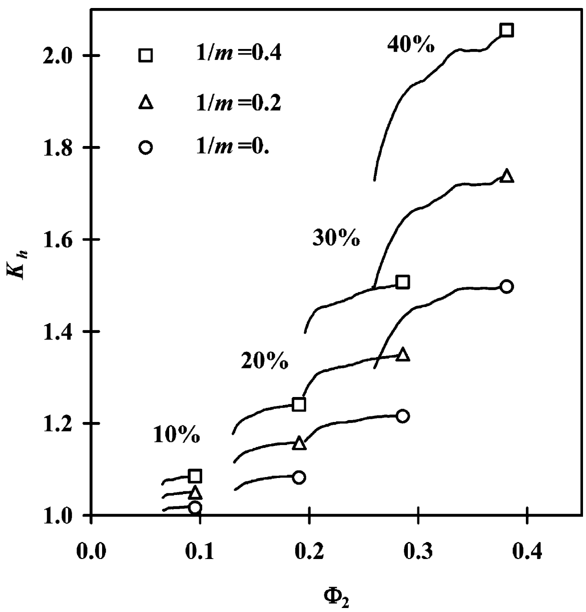

Fig. 9. Effect of the hardening coefficient m on the evolution of the constraint factor K_h in hydrostatic condition when all contacts are set with a 0.1 friction coefficient. Φ_2 is the volume fraction of hard particles relative to the total volume of the compact (Eq. (15)). See Table 1 for comparison with experimental data.

Experimental data concerning the constraint factor in isostatic conditions are not numerous and cannot always be easily compared to our numerical simulations that use monosize mixtures and spherical particles. Also, differences in initial density may affect the comparison with experimental data. With these restrictions in mind, we briefly review some of the available experimental data by collecting the value of K_h for two volume fractions of hard particles (20 and 40 vol.%) in Table 1. We restrict experimental data to systems for which the soft phase can be considered spherical, and for which the hard phase can be considered elastic. The value of K_h is taken in between 0.80 and 0.85 relative density at which it is believed that the main assumptions of the model are valid. The experimental data give consistent values for a value of K_h of approximately 1.3 at 20% fraction and 1.8 at 40% fraction of hard particles. These values are in reasonably good agreement with the curves shown in Fig. 9, for which all contacts are set with a friction coefficient of 0.1.

5.2. Close die compaction

In this section, we investigate the effect of a non-fully hydrostatic stress state on the response of the composite. We define K_{cd} as the ratio of the axial stress for the composite to the axial stress for the homogeneous material at the same relative density. We have verified that the value of the con-

Table 1
Review of experimental data for the value of K_h at approximately 0.80–0.85 relative density for different composite systems

Reference	Soft powder	$1/m$	Hard powder	Shape	D_0	K_h 20 vol.%	K_h 40 vol.%
Turner and Ashby [3]	plasticene	n.a.	glass	spherical	0.62	1.3	1.7
Kim et al. [11]	Cu	0.24	W	irregular	0.63	1.3	2.0
Sridhar and Fleck [14]	Al	0.24	Sic	irregular	0.63	n.a.	1.75
Sridhar and Fleck [14]	Pb	0.21	steel	spherical	0.63	1.2	1.7

straint factor is not significantly affected if another definition is adopted (for example, if the mean pressure is taken instead of the axial stress). Fig. 10 shows that the evolution of K_{cd} differs somewhat from the evolution of K_h as compaction proceeds. The value of K_{cd} is approximately constant during compaction for particle fractions smaller than 40 vol.%. This contrasts with the compaction behaviour in isostatic conditions for which K_h is increasing. Thus the resulting value of K_{cd} is

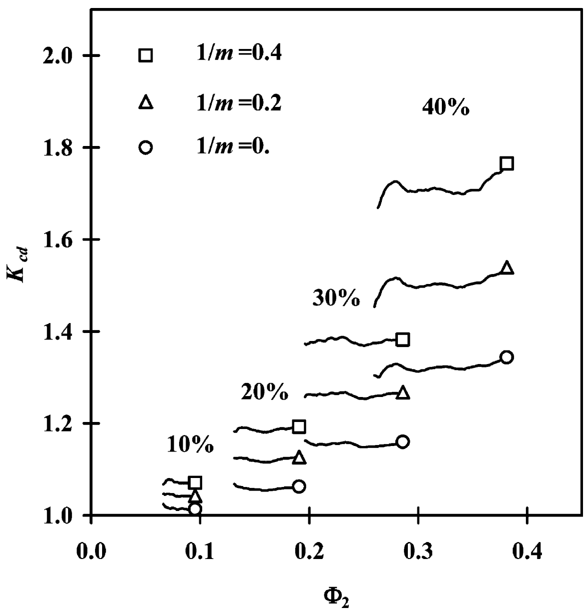

Fig. 10. Evolution of the constraint factor K_{cd} in close die conditions for the case where all contacts are set with a 0.1 friction coefficient and for different values of the hardening coefficient m. Φ_2 is the volume fraction of hard particles relative to the total volume of the compact (Eq. (15)). See Table 2 for comparison with experimental data.

smaller compared to K_h for a given relative density. The reason for this difference in behaviour is principally linked to the ratio N_{22}/N_{11}, which is smaller for close die compared to its value in isostatic compaction (Fig. 6). We have already observed that close die conditions induce an additional rearrangement of particles for homogeneous compacts compared to isostatic conditions [16]. The smaller value of the ratio N_{22}/N_{11} in close die conditions is presumably associated to the additional rearrangement brought by deviatoric strains.

Again, keeping in mind the restrictions that are inherent to the comparison of experimental data to the idealised model presented here, we collect in Table 2 experimental data for close die compaction. The same conditions and definitions apply for the data collected for close die as for isostatic compaction. It is interesting to note that despite the diversity of powders, the values of K_{cd} are quite similar from one composite to another, except for the data collected by Sridhar and Fleck [14]. These authors measured slightly lower constraint factors for an Al–SiC composite system compared to other authors. Apart from this last set of experimental data, the value of K_{cd} is consistent and is approximately 1.3 at 20% fraction and 1.6 at 40% fraction of hard particles. Again the model compares reasonably well with the experimental data for hardening exponent in between 0.2 and 0.4 and a friction coefficient set to 0.1. Concerning the difference in compaction behaviour that the model predicts between isostatic and close die compaction, the experimental data do not allow a positive conclusion. However, if one compares the value collected both in isostatic (Table 1) and close die (Table 2) compaction from the same study

Table 2
Review of experimental data for the value of K_{cd} (close die compaction) at approximately 0.80–0.85 relative density for different composite systems. Note that Lange et al. [1] have a different definition for the constraint factor using the *relative matrix density* for comparing pressures between the composite and the homogeneous system (Fig. 4 in [1]). The table reports values of K_{cd} that are consistent with the definition adopted in the present article

Reference	Soft powder	$1/m$	Hard powder	Shape	D_0	K_{cd}	K_{cd}
						20 vol.%	40 vol.%
Lange et al. [1]	Al	n.a.	steel	spherical	0.64	≈1.2	1.6
Lange et al. [1]	Pb	n.a.	steel	spherical	0.6	≈1.2	1.6
Kim et al. [11]	Cu	0.24	W	irregular	0.63	1.3	1.7
Sridhar and Fleck [14]	Al	0.24	Sic	irregular	0.63	n.a.	1.3
Martin et al. [15]	Al	0.24	W_2C	irregular	0.55	1.3	1.6

[11,14], the value of K_{cd} is consistently smaller than or equal to K_h.

In close die compaction, additional macroscopic information is given by the ratio between the transverse stress and the axial stress σ_{11}/σ_{33}. Fig. 11 shows the evolution of the ratio σ_{11}/σ_{33} with relative density for the composites and the homogeneous compact for a ideally plastic soft phase and with all contacts characterised by a 0.1 friction coefficient. Fig. 11 shows that there is only a slight effect on the ratio when adding hard particles (increase of the ratio with an increase of the fraction of hard particles). Also, we have verified that the evolution of the ratio is only very slightly dependent on the hardening coefficient (decrease of the ratio with an increase of $1/m$). The evolution of the ratio σ_{11}/σ_{33} consists first in a decrease at low relative density (due to the fact that the compact is initially isotropic) followed by an increase. In a precedent study [16], we have shown that the range of experimental value for this ratio is quite wide for homogeneous powders, spanning from 0.30 to 0.75 [14,30–32], so it is difficult to compare the available experimental data to the model. The experimental data indicate an increase of σ_{11}/σ_{33} as densification proceeds which is in accordance with the model [16].

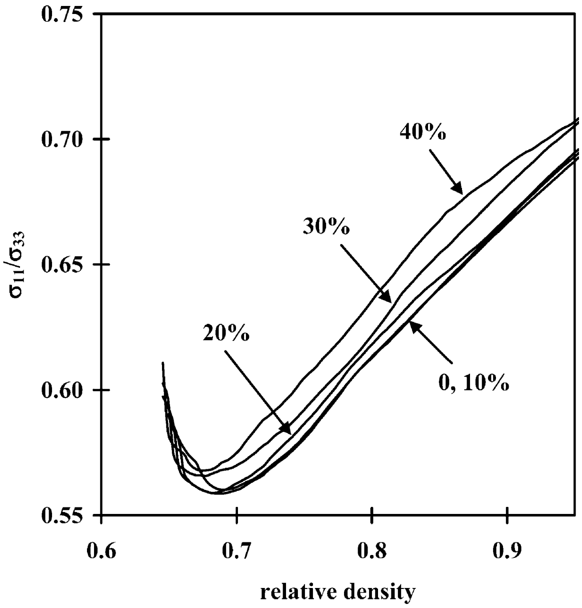

Fig. 11. Evolution of the transverse to axial stress ratio (σ_{11}/σ_{33}) in close die conditions for the case where all contacts are set with a 0.1 friction coefficient. Soft particles are ideally plastic ($1/m = 0$).

6. Conclusion

The compaction of monosize mixtures of soft and hard particles has been studied by using the DEM. Two main mechanisms have generally been proposed to account for the retardation of compaction, namely the effect of a network of hard particles that supports a larger portion of the load and the additional deformation that soft particles must undergo in the presence of hard particles [1]. These two mechanisms have been observed with the numerical simulations presented here. We have shown that the friction coefficient of the hard–hard particle contacts plays a significant role in the development of a mechanical network of hard particles. In other words, if the rearrangement of hard particles is restrained by highly frictional contacts, the network carries a larger than average portion of the load, thus limiting the deformation of the soft phase and retarding the densification of the compact. We have limited the present study to three idealized friction conditions: frictionless contacts, frictionless hard–hard contacts and frictional contacts for all type of contacts, thus demonstrating the effect of local rearrangement. The actual measurement of a friction coefficient is a difficult task for metallic and ceramic powders and the addition of lubricant in industrial conditions certainly has a strong effect on the friction between particles. However, the present simulations that use a realistic friction coefficient provide a reasonable match with experimental data concerning the constraint factor. Apart from friction, another barrier to rearrangement is the shape of particles. The idealized situation of a mixture of spherical particles is far from a real mixture containing hard ceramic angular particles or crushed irregularly shaped particles. The DEM permits the introduction of non-spherical particles but to the cost of large CPU time. An alternative is to keep spherical particles and to introduce a high friction coefficient in order to reduce rearrangement in these mixtures. Another practical situation that would introduce larger constraints between hard particles would be the case of partially sintered hard particles. In that case, the effect of the hard inclusions would be much more dramatic than in the present study.

The additional deformation that soft particles must undergo in the presence of hard particles has been demonstrated by measuring the additional indentation sustained by soft particles at the contact with hard particles. A clear signature of the effect of the presence of hard particles is the strong effect of the hardening coefficient *m* on the compaction curve. Hence a quantitative study of the compaction of mixtures requires the introduction of hardening effects in the constitutive equation of the soft phase.

References

[1] Lange FF, Atteraas L, Zok F, Porter JR. Acta metall. mater. 1991;39:209.
[2] Bouvard D, Lange FF. Acta metall. mater. 1991;39:3083.
[3] Turner CD, Ashby MF. Acta mater. 1996;44:4521.
[4] Bouvard D. Powder Technol. 2000;111:231.
[5] Zavaliangos A, Laptev A. Acta mater. 2000;48:2565.
[6] Arzt E. Acta metall. mater. 1982;30:1883.
[7] Bouvard D. Acta metall. mater. 1993;41:1413.
[8] Zavaliangos A, Lam A, Wen J. In: Cadle TM, Narasimhan KS, editors. Advances in powder metallurgy & particulate materials, vol. 16. Washington DC: MPIF; 1996. p. 13-7.
[9] Storåkers B, Fleck NA, McMeeking RM. J. Mech. Phys. Solids 1999;47:785.
[10] Fleck NA. J. Mech. Phys. Solids 1995;43:1409.
[11] Kim KT, Cho JH, Kim JS. J. Eng. Mater. Tech. 2000;122:119.
[12] Jagota A, Scherer GW. J. Am. Ceram. Soc. 1993;76:3123.
[13] Jagota A, Scherer GW. J. Am. Ceram. Soc. 1995;78:521.
[14] Sridhar I, Fleck NA. Acta mater. 2000;48:3341.
[15] Martin CL, Lame O, Bouvard D. Mech. Mater. 2000;32:405.
[16] Martin CL, Bouvard D, Shima S. J. Mech. Phys. Solids (in press).
[17] Skrinjar O, Larsson P-L. Discrete element modelling of cold compaction of composite powders. In: Proceedings of World Congress on Powder Metallurgy & Particulate Materials (PM^{2}TEC 2002)/. Orlando, USA: MPIF (in press).
[18] Storåkers B, Biwa S, Larsson PL. Int. J. Solids Struct. 1997;34:3061.
[19] Carlsson S, Larsson L, Biwa S. Int. J. Mech. Sc. 2000;42:107.
[20] Larsson J, Storåkers B. J. Mech. Phys. Solids 2002;50:2029.
[21] Redanz P, Fleck NA. Acta mater. 2001;49:4325.
[22] Cundall PA, Strack A. Géotechnique 1979;29:47.
[23] Thornton C, Antony SJ. Phil. Trans. R. Soc. London A 1998;356:2763.
[24] Christoffersen J, Mehrabadi MM, Nemat-Nasser S. J. Appl. Mech. 1981;48:339.
[25] Dodds JA. J. Coll. Interf. Sci. 1980;77:317.
[26] Helle AS, Easterling KE, Ashby MF. Acta metall. 1985;33:2163.
[27] Mason G. Nature 1968;217:733.
[28] Finney JL. Proc. Royal Soc. London A 1970;319:479.
[29] Yen KZY, Chaki TK. J. appl. Phys. 1992;71:3164.
[30] Brown SB, Abou-Chedid G. J. Mech. Phys. Solids 1994;42:383.
[31] Watson TJ, Wert JA. Metall. Trans. 1993;24A:2071.
[32] Mosbah P, Bouvard D, Ouedraougo E, Stutz P. Powder Metall. 1997;40:269.